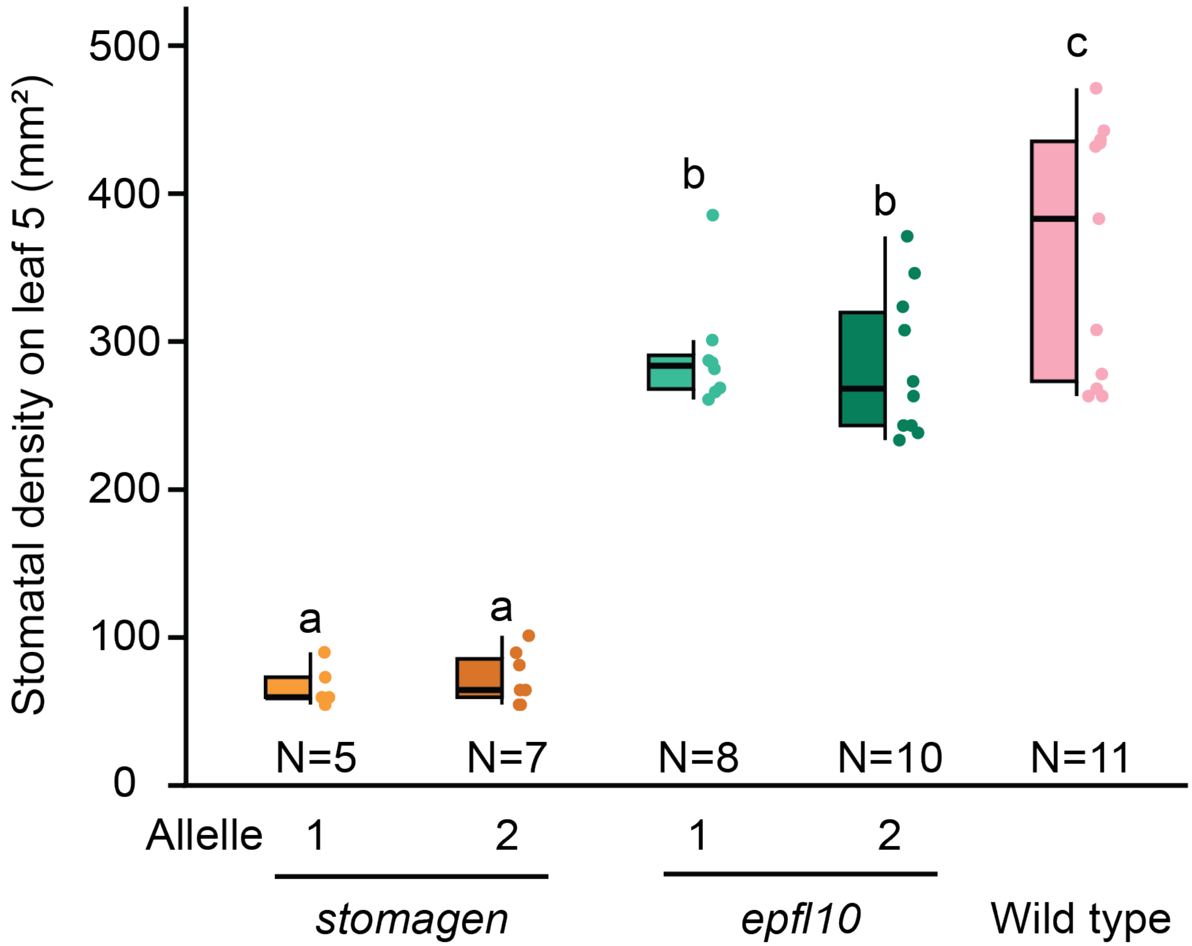
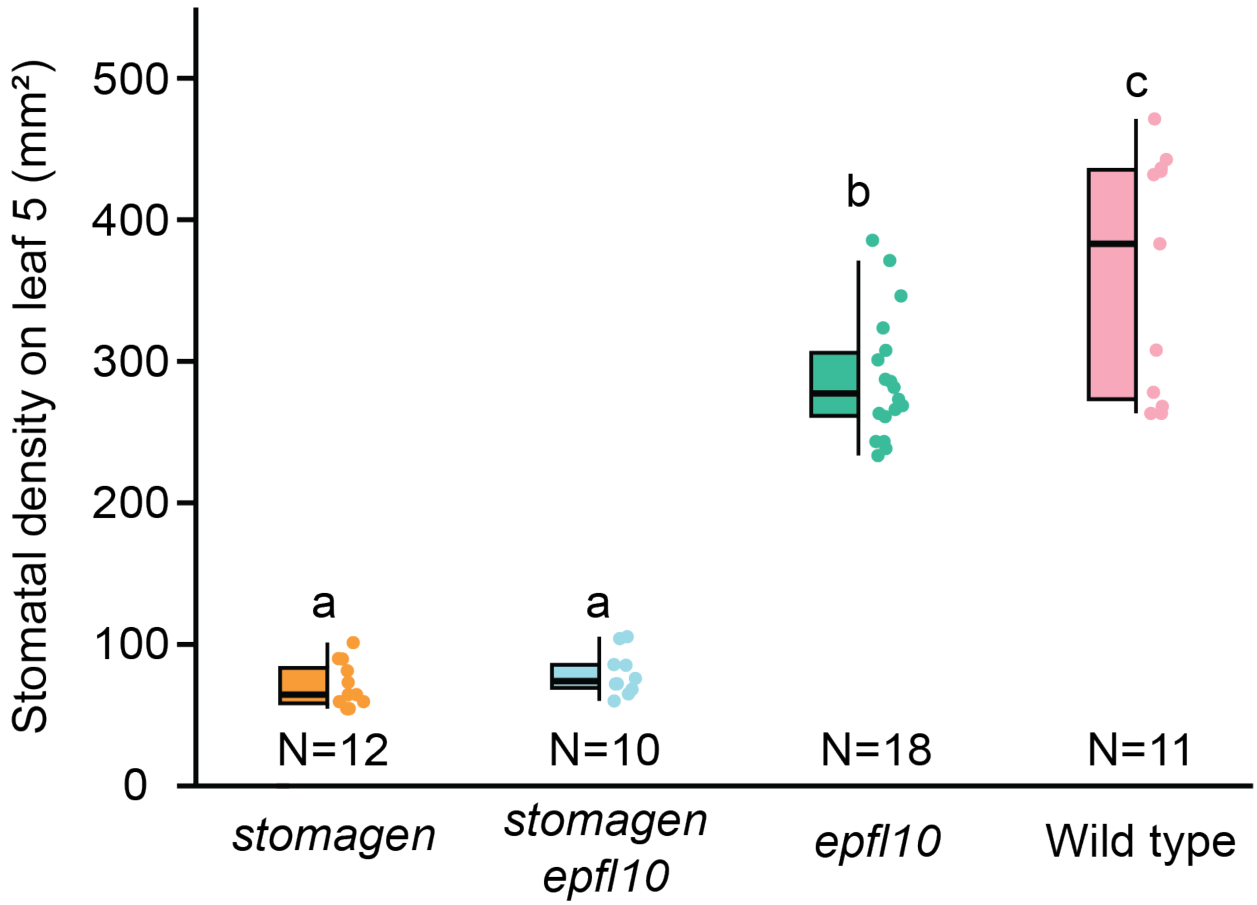
Supplemental Figure S1. Leaf 5 stomatal density of each *stomagen* and *epfl10* allele



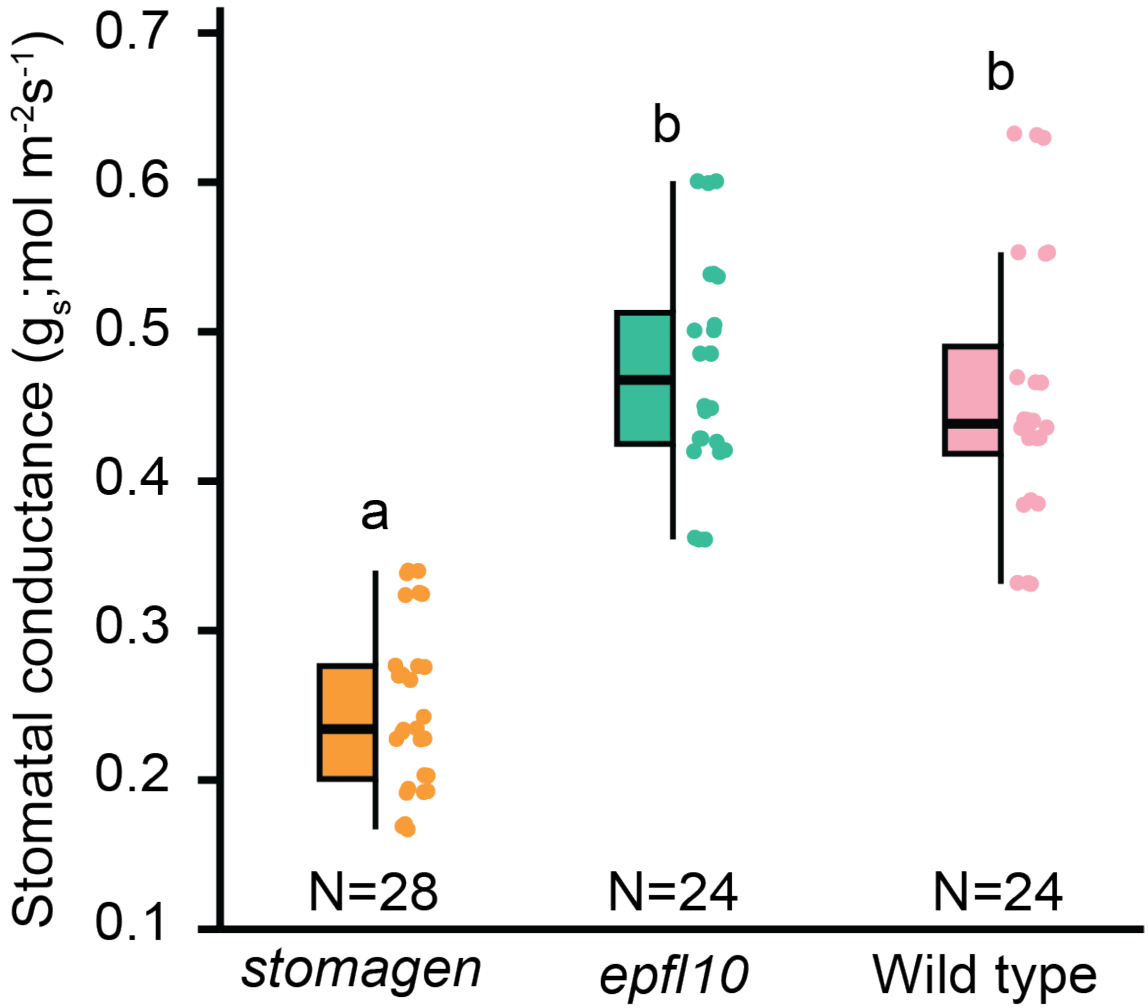
Stomatal density of the fifth fully expanded true leaf. Stomatal density measurements of each allele of *stomagen; epfl10;* and wild type taken on 21-day old plants grown in the growth chamber. Allele numbers correspond to the alleles described in Figure 1E,F. In this box-and-whisker plot the center horizontal indicates the median, upper and lower edges of the box are the upper and lower quartiles and whiskers extend to the maximum and minimum values within 1.5 interquartile ranges. Letters indicate a significant difference between means ($P<0.05$, one-way ANOVA Tukey HSD post-hoc test).

Supplemental Figure S2. Leaf 5 stomatal density of *stomagen epfl10* double knockout



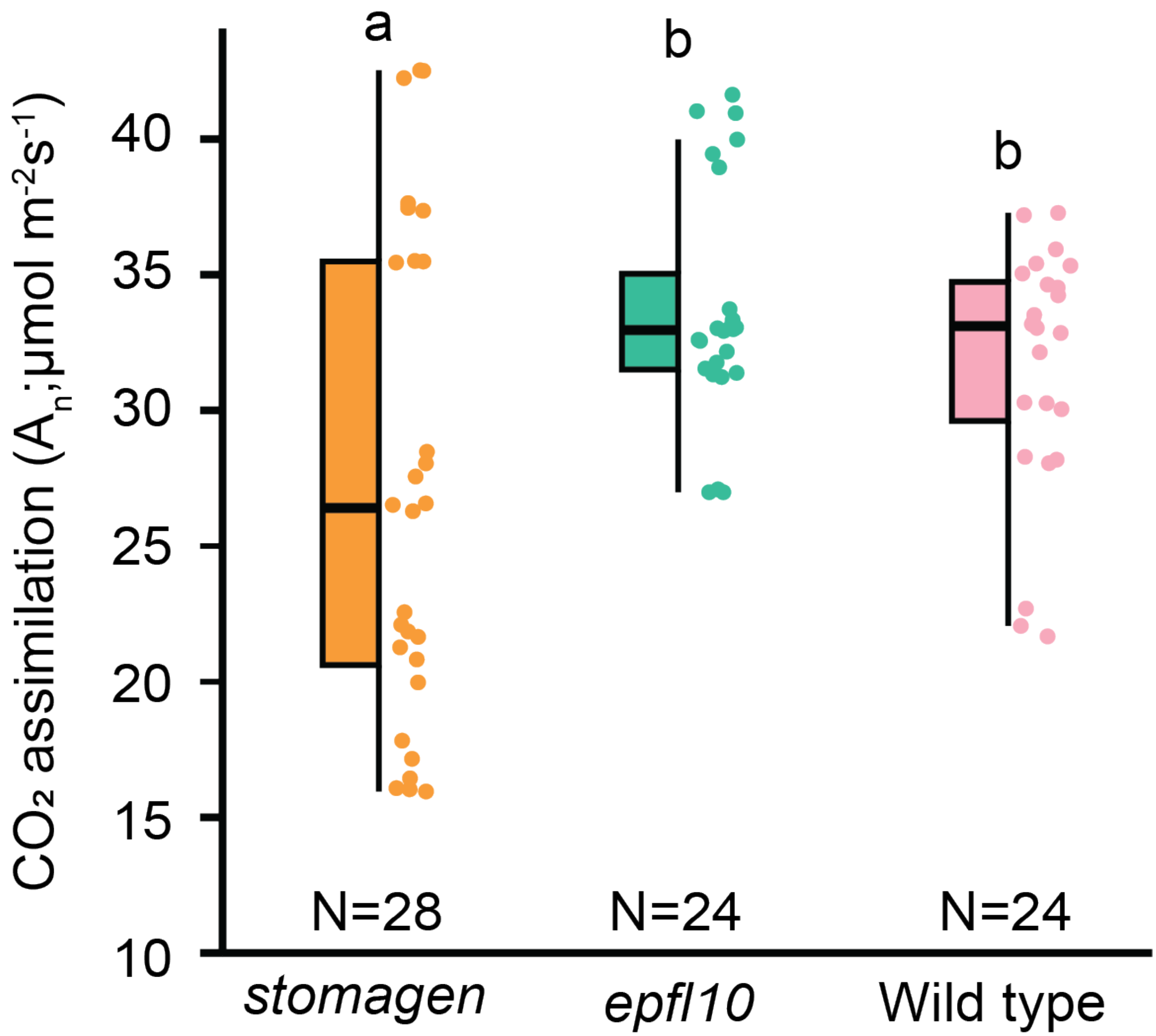
Stomatal density of the fifth fully expanded true leaf. Stomatal density measurements of *stomagen, epfl10; stomagen; epfl10;* and wild type taken on 21-day old plants grown in the growth chamber. In this box-and-whisker plot the center horizontal indicates the median, upper and lower edges of the box are the upper and lower quartiles and whiskers extend to the maximum and minimum values within 1.5 interquartile ranges. Letters indicate a significant difference between means ($P<0.05$, one-way ANOVA Tukey HSD post-hoc test).

Supplemental Figure S3. Growth chamber stomatal conductance measurements at 1000 µmol photons m^{-2} s^{-1}



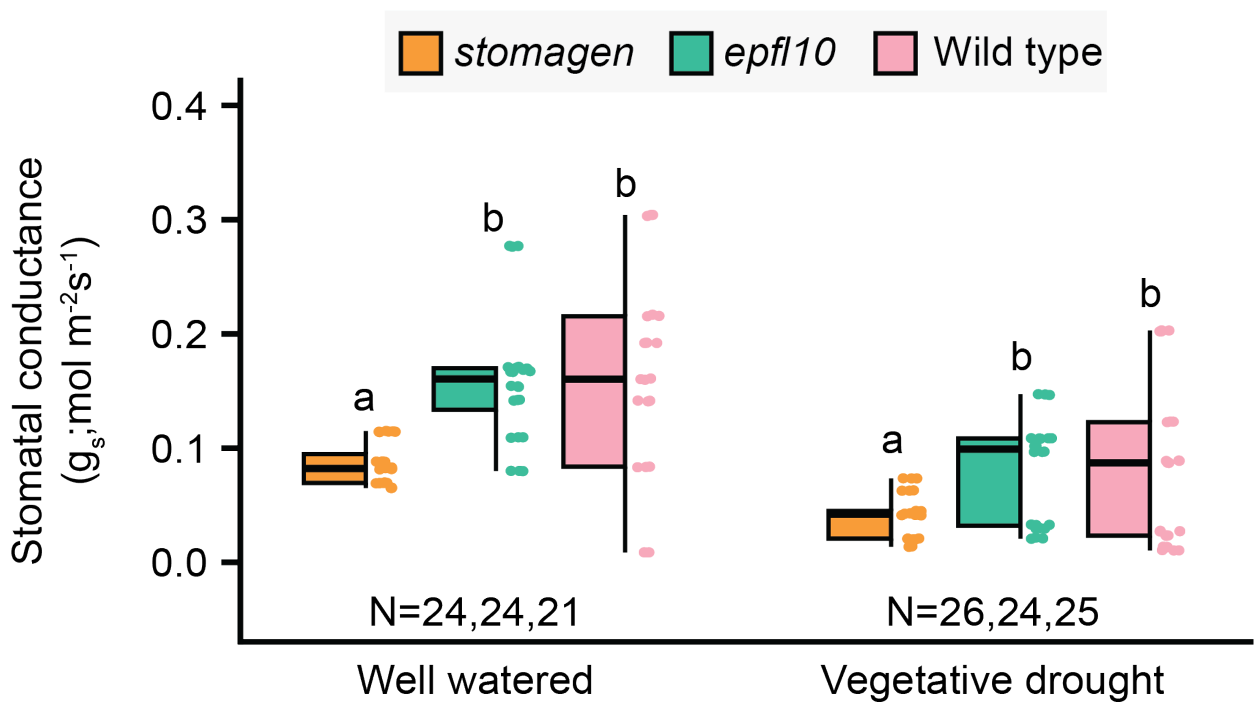
Stomatal conductance measurements of *stomagen*, *epfl10*, and wild type at 1000 µmol photons m^{-2} s^{-1} grown in the growth chamber. In this box-and-whisker plots the center horizontal indicates the median, upper and lower edges of the box are the upper and lower quartiles and whiskers extend to the maximum and minimum values within 1.5 interquartile ranges. Letters indicate a significant difference between means ($P<0.05$, one-way ANOVA Tukey HSD post-hoc test).

Supplemental Figure S4. Growth chamber carbon assimilation measurements at 1000 µmol photons m^{-2} s^{-1}



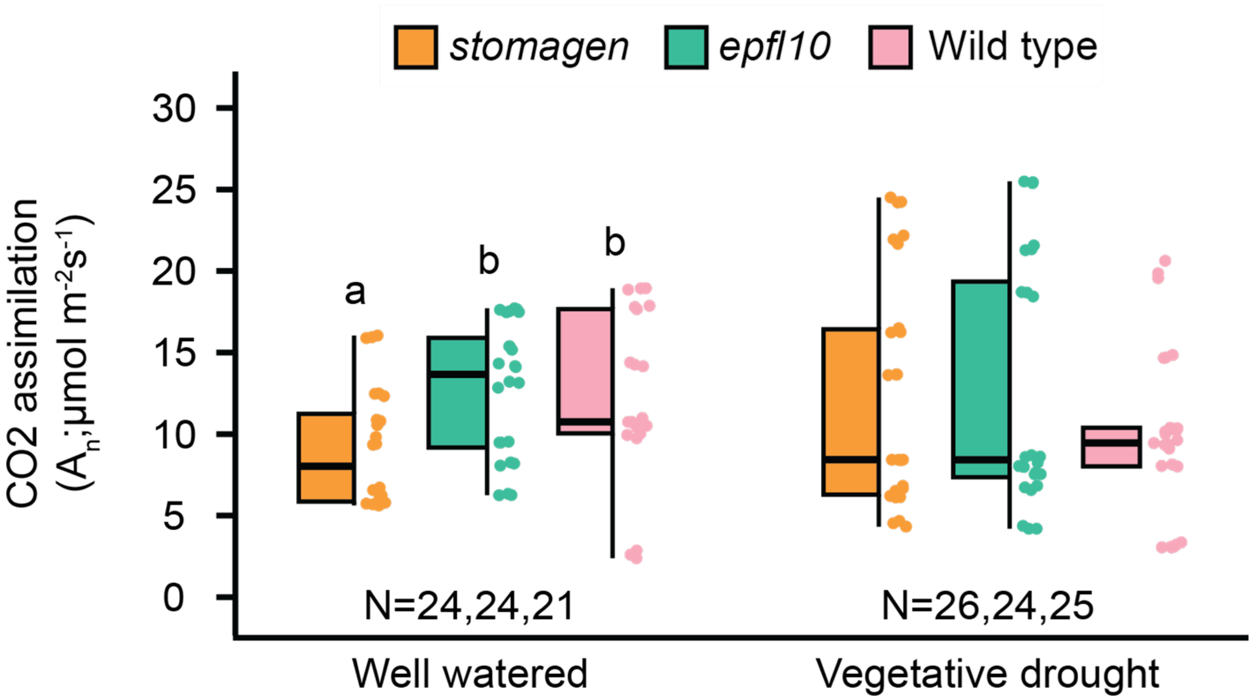
Carbon assimilation measurements of *stomagen*, *epfl10*, and wild type at 1000 µmol photons m^{-2} s^{-1} grown in the growth chamber. In this box-and-whisker plot the center horizontal indicates the median, upper and lower edges of the box are the upper and lower quartiles and whiskers extend to the maximum and minimum values within 1.5 interquartile ranges. Letters indicate a significant difference between means ($P<0.05$, one-way ANOVA Tukey HSD post-hoc test). Measurements were taken on full expanded leaf 5 of 28-day-old plants grown in well-watered conditions plants using an infrared gas analyzer (LI6800XT, LI-COR, Lincoln, NE, USA) with chamber conditions set to: light intensity 1000 µmol photons m^{-2} s^{-1}(90% red light, 10% blue light); leaf temperature 27°C; flow rate 500 µmol s^{-1}; relative humidity 40%; and CO_2 concentration of sample 400 µmol mol^{-1}.

Supplemental Figure S5. Growth chamber carbon assimilation measurements at 1000 µmol photons m^{-2} s^{-1}



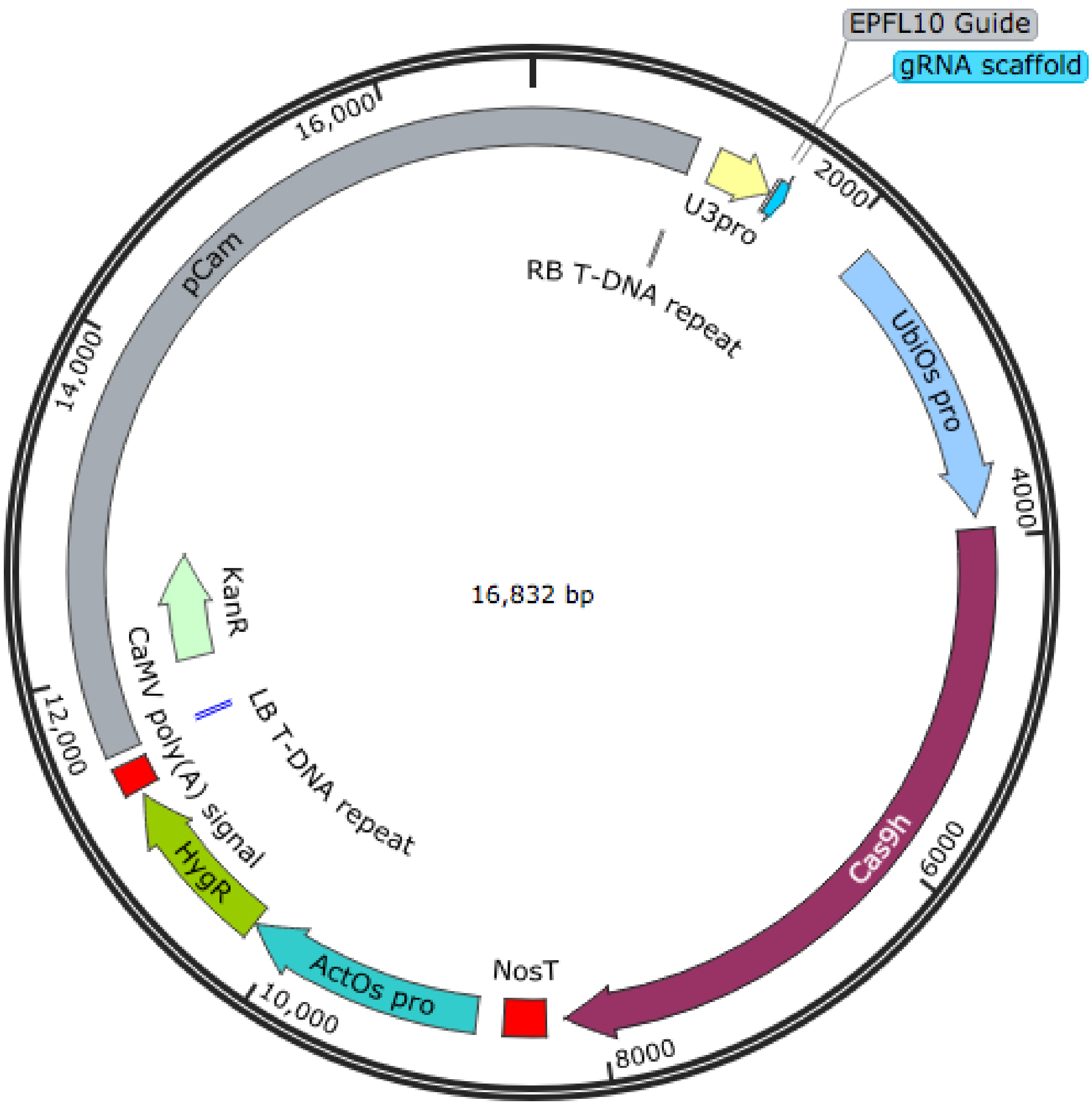
Measurements of stomatal conductance of *epfl10* and *stomagen* lines in Nipponbare (*Oryza sativa* spp. Japonica) grown in the greenhouse under two watering regimes: well-watered, and vegetative drought. In this box-and-whisker plot the center horizontal indicates the median, upper and lower edges of the box are the upper and lower quartiles and whiskers extend to the maximum and minimum values within 1.5 interquartile ranges. Outliers are represented by black dots. Letters indicate a significant difference between means ($P<0.05$, one-way ANOVA Tukey HSD post-hoc test).

Supplemental Figure S6. Greenhouse carbon assimilation measurements at 1000 µmol photons m^{-2} s^{-1} under two watering regimes



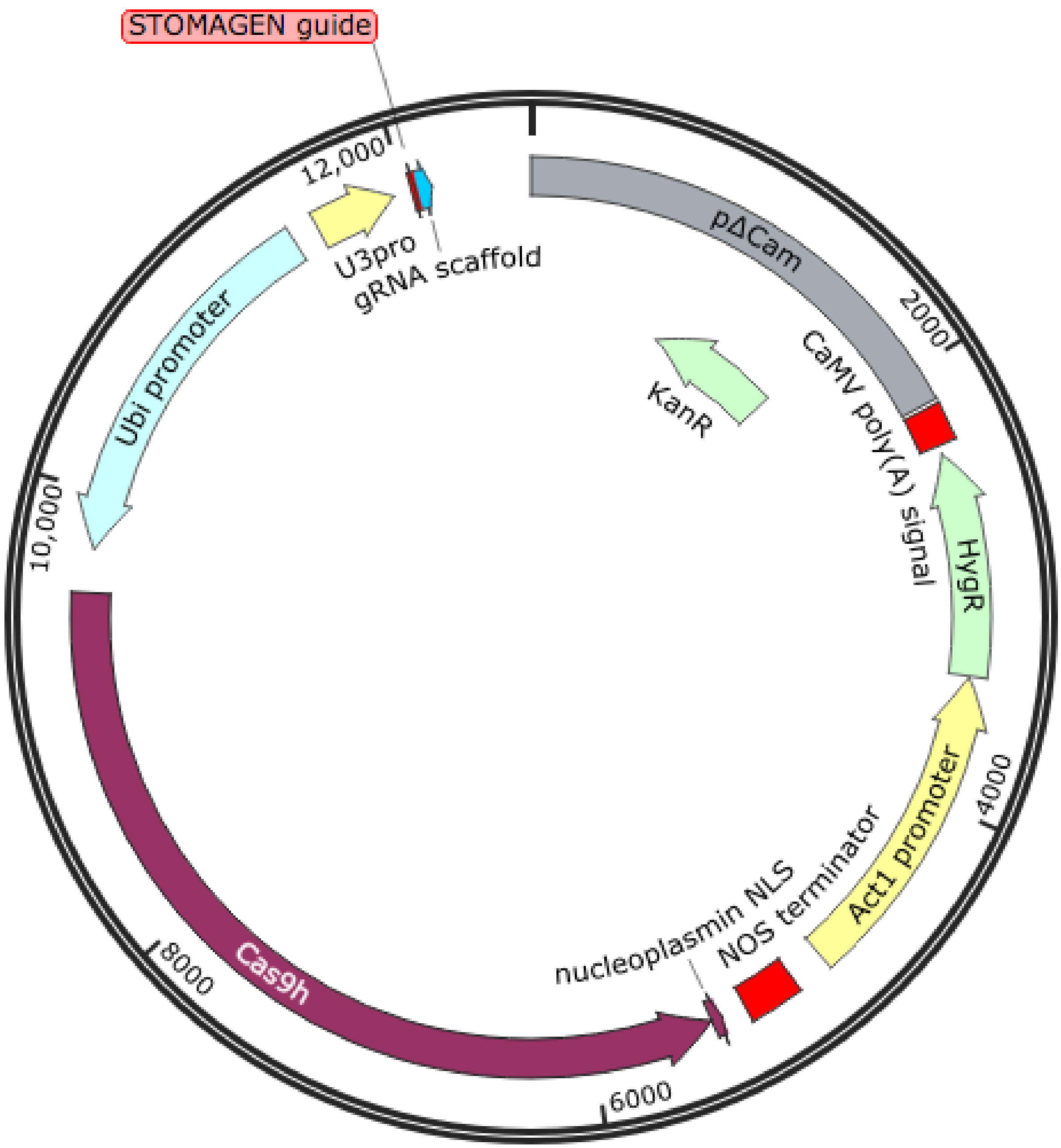
Measurements of carbon assimilation *epfl10* and *stomagen* lines in Nipponbare (*Oryza sativa* spp. Japonica) grown in the greenhouse under two watering regimes: well-watered, and vegetative drought. In this box-and-whisker plot the center horizontal indicates the median, upper and lower edges of the box are the upper and lower quartiles and whiskers extend to the maximum and minimum values within 1.5 interquartile ranges. Letters indicate a significant difference between means ($P<0.05$, one-way ANOVA Tukey HSD post-hoc test). Measurements were taken on full expanded leaf 5 of 28-day-old plants grown in well-watered conditions and vegetative drought.

Supplemental Figure S7. OsEPFL10 editing vector map



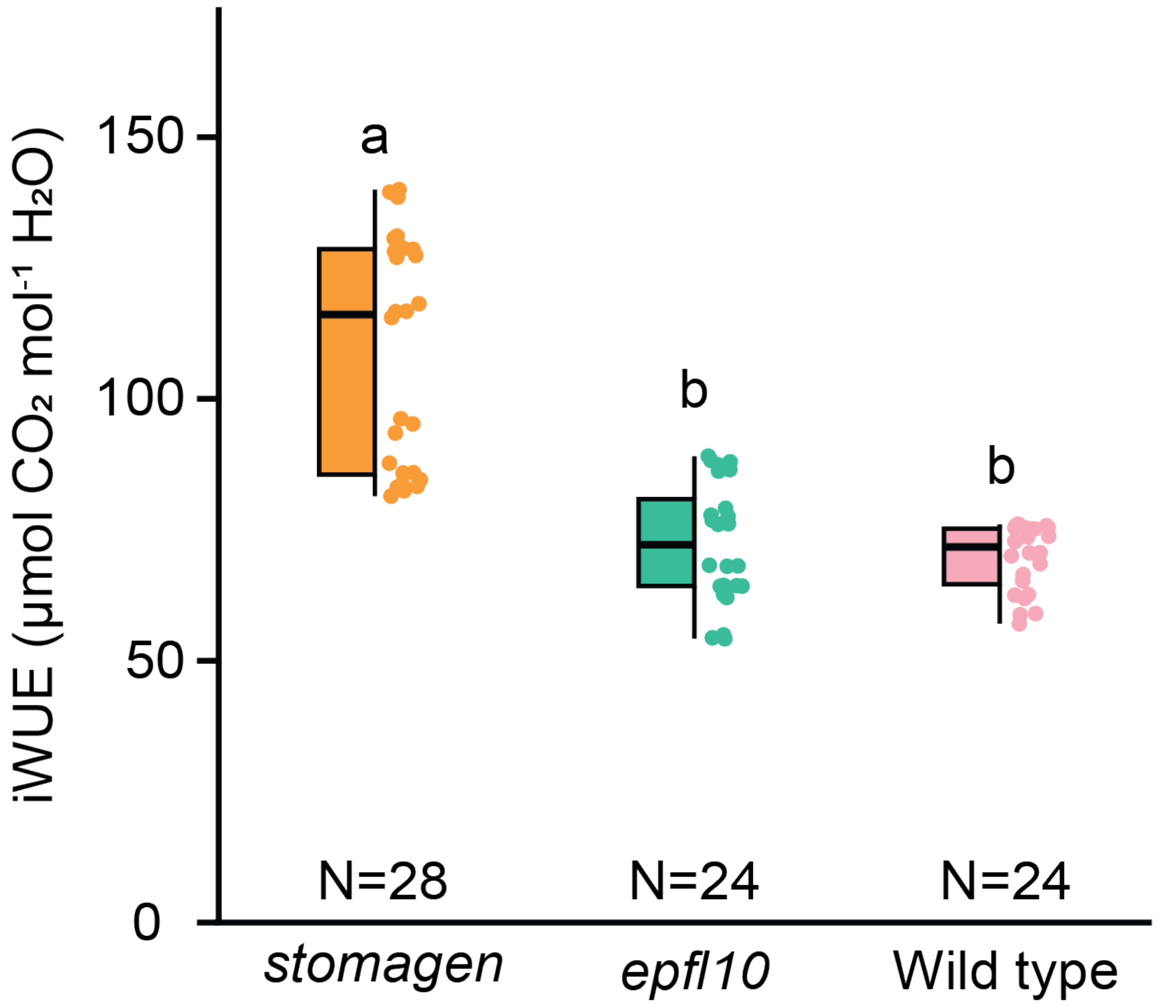
Representative plasmid map of binary gene editing vector used to generate *OsEPFL10* mutations. Full plasmid sequence is available in supporting information.

Supplemental Figure S8. OsSTOMAGEN editing vector map



Representative plasmid map of gene editing vector used to generate *OsSTOMAGEN* mutations. Full plasmid sequence is available in supporting information.

Supplemental Figure S9. Growth chamber iWUE efficiency measurements at 1000 µmol photons m^{-2} s^{-1}



Intrinsic water-use efficiency of *epfl10* and *stomagen* lines in Nipponbare (*Oryza sativa* spp. Japonica) grown in the growth chamber. In this box-and-whisker plots the center horizontal indicates the median, upper and lower edges of the box are the upper and lower quartiles and whiskers extend to the maximum and minimum values within 1.5 interquartile ranges. Letters indicate a significant difference between means ($P<0.05$, one-way ANOVA Tukey HSD post-hoc test). Measurements were taken on full expanded leaf 5 of 28-day-old plants grown in well-watered conditions. Intrinsic water use efficiency (iWUE) is calculated by dividing photosynthesis by stomatal conductance for each biological replicate.